
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Out-of-distribution (OOD) detection is fundamental to the safe and reliable de-
2 ployment of machine learning systems in open-world environments, where models
3 routinely encounter inputs drawn from classes absent at training time. A promising
4 line of work leverages unlabeled in-the-wild data to sharpen OOD detection, yet
5 such data is intrinsically difficult to exploit: it constitutes an unlabeled mixture of
6 in-distribution (InD) and OOD samples, and the absence of any cleanly delineated
7 set of OOD examples obstructs the training of an effective OOD classifier. We
8 introduce Medix, a framework that resolves this difficulty by identifying candidate
9 outliers directly from unlabeled data through the median operation. We adopt the
10 median precisely because, as a robust estimator of central tendency, it is prov-
11 ably insensitive to the noise and OOD contamination that pervade wild data. The
12 outliers it isolates, together with labeled InD data, are then used to train a robust
13 OOD classifier. On the theoretical side, we derive error bounds establishing that
14 Medix attains a low misclassification rate, and our experiments corroborate these
15 guarantees: across open-world settings, Medix consistently outperforms existing
16 methods, confirming the validity of our theoretical analysis.

17 1 Introduction

18 The deployment of machine learning models in real-world applications inevitably exposes them to
19 out-of-distribution (OOD) inputs, posing acute challenges for safety and reliability. Such inputs,
20 originating from categories the model was never trained to recognize, ought not to receive confident
21 predictions; yet neural networks are inherently ill-equipped for this task, typically lacking any
22 mechanism to detect and appropriately abstain on OOD inputs at deployment time (Nguyen et al.,
23 2015).

24 Identifying OOD samples at inference is both critical and intrinsically difficult: because models are
25 never exposed to unknown distributions during training, they cannot, on their own, reliably separate
26 OOD from in-distribution (InD) data. To confront this challenge, a recent line of work (Katz-Samuels
27 et al., 2022a; Du et al., 2024a) has turned to auxiliary “in-the-wild” data. In particular, Katz-Samuels
28 et al. (2022a) regularize model training with unlabeled wild data while retaining the objective of
29 classifying labeled InD data. The appeal of wild data lies in its availability—it is readily collected
30 once a model is operating in its target environment—and in the fact that it lets the model absorb the
31 true distribution of OOD inputs encountered at test time, yielding substantially more robust OOD
32 detection.

33 Exploiting unlabeled wild data, however, is far from straightforward, precisely because it arrives as an
34 entangled mixture of InD and OOD data. The absence of any clean, delineated set of OOD samples
35 complicates the design of robust detectors, since the OOD detector observes only samples drawn
36 from this mixed distribution and is never told whether a given sample is InD or OOD. This regime
37 remains substantially underexplored, leaving considerable room for advancement. Compounding

the gap, few studies furnish a formal theoretical foundation; to the best of our knowledge, Du et al. (2024a) stands as the sole prior work to provide such a foundation for the “in-the-wild” setting.

At the same time, median-based approaches have recently proven highly effective in data pruning (Acharya et al., 2024). Motivated by these two threads—the open problem of exploiting wild data and the demonstrated robustness of median-based estimation—this paper sets out to answer the following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question in the affirmative by introducing a median-centric perspective on OOD detection. Concretely, we formulate a median-based optimization framework and develop an accompanying algorithm, Medix, that identifies OOD samples within wild data at a demonstrably low error rate. We establish a theoretical foundation that guarantees this low error, and we validate it empirically, exhibiting both the robustness and the efficiency of the algorithm. In doing so, we contribute one of the few rigorous theoretical foundations for the unlabeled “in-the-wild” setting. We prove that median-based filtering is robust for outlier detection in unlabeled mixtures by bounding the fraction of InD samples erroneously flagged as outliers. This fraction is governed by two complementary effects: a *contamination effect*, which quantifies the influence of OOD points and remains controlled whenever the OOD proportion stays below 50%; and a *concentration effect*, which exploits the sub-Gaussian nature of InD gradients to bound the deviation of individual InD points from their mean gradient. Jointly, these effects guarantee that, with high probability, the number of misclassified InD points remains small—establishing the method’s reliability even under worst-case contamination.

We benchmark Medix against two families of methods: (1) those trained solely on InD data, and (2) those trained on both InD data and an auxiliary unlabeled dataset. On CIFAR-100 (Krizhevsky et al., 2009), Medix improves markedly over the strong baseline KNN+ (Sun et al., 2022), lowering average FPR95 by 40.98%. Crucially, whereas approaches such as Outlier Exposure (Hendrycks et al., 2019) depend on a clean auxiliary dataset—imposing the strong distributional assumption that the auxiliary data is fully separable from the InD data—Medix attains superior results without any such assumption, and is therefore far more broadly applicable. Relative to WOODS (Katz-Samuels et al., 2022a), Medix further reduces average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10.

We summarize our key contributions as follows:

- C1) A median-centric filtering framework.** We propose Medix, a median-centric greedy method that first filters outliers from unlabeled wild data and then trains an OOD detector on the extracted outliers together with the labeled InD samples. The filtering stage constitutes our central methodological contribution.
- C2) Two-sided theoretical guarantees.** We establish provable guarantees for the robustness of median-based filtering in jointly identifying inliers and outliers within unlabeled mixtures. Specifically, we prove that the misclassification rates for both InD samples flagged as outliers and OOD samples retained as inliers are tightly controlled by two effects: a contamination effect, which remains bounded whenever the OOD proportion is below 50%, and a concentration effect, which follows from the stable gradient behavior of InD samples.
- C3) Comprehensive empirical validation.** We conduct an extensive evaluation of Medix across eleven InD–OOD pairs against 20 competitive baselines, and Medix outperforms every one of them, delivering superior performance across the board. We further show that Medix outlier extraction attains a low error rate (e.g., only 12.5%; see Figure 2), directly corroborating our theoretical findings.

2 Preliminaries and Problem Setup

In this section we formalize the OOD detection problem and then specify the data setup, model architecture, loss functions, and learning objective that underpin the remainder of the paper.

Labeled In-Distribution Data. Let $\mathcal{X}$ denote the input space and $\mathcal{Y} = \{1, \dots, K\}$ the label space; together they characterize the structure of the InD data. A labeled dataset $\mathcal{S}_{\text{in}} =$

87 $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ comprises n pairs drawn independently and identically distributed (i.i.d.)
 88 from $\mathbb{P}_{XY}$, an unknown joint distribution over $\mathcal{X} \times \mathcal{Y}$. We write $\mathbb{P}_{\text{in}}$ for the marginal of $\mathbb{P}_{XY}$ on $\mathcal{X}$,
 89 i.e., the underlying distribution of InD inputs, and we train the InD model on $\mathcal{S}_{\text{in}}$.

90 **Out-of-distribution detection.** We consider the practical scenario in which a model, trained on
 91 labeled InD data, is subsequently deployed in environments that may present OOD inputs from
 92 classes unseen during training, i.e., inputs carrying some label $y \notin \mathcal{Y}$. On such inputs the model must
 93 abstain rather than commit to a class prediction. The central objective at inference time is thus to
 94 decide whether a given input is drawn from the InD distribution or originates from an OOD source.

95 **Unlabeled wild data.** A principal obstacle in OOD detection is the scarcity of labeled OOD samples:
 96 the OOD sample space is potentially vast, rendering the collection of labeled examples both costly
 97 and impractical. To circumvent this, and following Katz-Samuels et al. (2022a), we incorporate
 98 unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into our learning framework so as to faithfully reflect
 99 real-world deployment. Wild data is a blend of InD and OOD samples and is readily gathered once
 100 a model pre-trained on $\mathcal{S}_{\text{in}}$ is put into operation. Consistent with Du et al. (2024a); Katz-Samuels
 101 et al. (2022a), we adopt the Huber contamination model (Huber, 1964) to characterize the marginal
 102 distribution of the wild data,

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

103 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution over $\mathcal{X}$. Note that the
 104 boundary case $\pi = 0$ corresponds to the absence of OOD samples and renders the problem trivial.

105 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier parameterized by $\phi \in \Phi$,
 106 with Φ its parameter space; the output of f_ϕ is a soft probability distribution over the $K = |\mathcal{Y}|$ InD
 107 classes, and its supervised loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection we introduce a separate
 108 classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$, parameterized by $\theta \in \Theta$ over parameter space Θ , whose associated binary
 109 loss is $\ell_b(g_\theta(\mathbf{x}), y_b)$ with $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$. Here the positive value $y_+ > 0$ encodes the InD
 110 class and the negative value $y_- < 0$ the OOD class, so that the sign of g_θ directly yields the detection
 111 decision.

112 **Learning objective.** Our framework jointly trains the OOD detector g_θ and the multi-class classifier
 113 f_ϕ , drawing simultaneously on the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$. At test time we assess
 114 performance with the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{XY}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

115 where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ is the OOD test distribution. FPR and TPR jointly
 116 characterize detection quality, while $\text{Acc}(f_\phi)$ certifies that InD classification is preserved.

117 3 Method: Median-Centric Framework for OOD Detection

118 We now present **Medix**, a learning paradigm that performs OOD detection by systematically har-
 119 nessing unlabeled wild data. By construction, Medix transcends the limitations of conventional
 120 approaches confined to InD data, and it is especially well-suited to open-world deployment, where
 121 models are routinely confronted with previously unseen inputs. The framework decomposes into
 122 two tightly coupled stages: **1) Outlier Extraction**, a filtering procedure that isolates candidate OOD
 123 samples from the unlabeled wild data (Section 3.1); and **2) Detector Training**, in which a binary
 124 OOD detector is trained on the InD data together with the outlier candidates identified in stage 1
 125 (Section 3.2), following the protocol of Du et al. (2024a). As the following sections establish, this
 126 two-stage design not only extracts OOD data from the unlabeled wild set effectively but also furnishes
 127 a robust foundation for deploying machine learning models in dynamic, open-world environments.

128 3.1 Extracting Candidate Outliers from the Wild Data

129 To isolate candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix adopts an optimization-based
 130 formulation grounded in the parameter gradients of a classification model f_ϕ trained exclusively on
 131 the InD dataset $\mathcal{S}_{\text{in}}$. The rationale is that gradients computed against an InD-trained model encode

how far a sample departs from InD behavior, providing a discriminative signal for filtering. We formalize the procedure below.

Reference gradient estimation from InD data. The first step is to establish a reference gradient from the InD dataset $\mathcal{S}_{\text{in}}$. We train the classifier f_ϕ on $\mathcal{S}_{\text{in}}$ via empirical risk minimization (ERM),

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. With the classifier trained, we define the reference gradient $\bar{\nabla}_{\text{in}}$ as the empirical mean, over the InD data, of the loss gradients with respect to the model parameters:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

The reference gradient $\bar{\nabla}_{\text{in}}$ thus serves as an anchor against which the deviation of any other data point can be quantified.

Motivation. The design of Medix rests on a testable hypothesis: as the proportion of OOD samples in the wild dataset $\mathcal{S}_{\text{wild}}$ grows, the aggregate gradient statistic of $\mathcal{S}_{\text{wild}}$ should deviate increasingly from the reference InD gradient $\bar{\nabla}_{\text{in}}$. To probe this hypothesis directly, we conduct a controlled experiment with CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. We initialize $\mathcal{S}_{\text{wild}}$ with 10,000 CIFAR-10 samples held disjoint from the data used to train $\phi_{\mathcal{S}_{\text{in}}}$ (and hence to compute $\bar{\nabla}_{\text{in}}$), then incrementally inject SVHN OOD samples while tracking the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the wild-set gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

As shown in Figure 1, the L_2 -norm deviation increases clearly and monotonically with the OOD count, confirming our hypothesis. This monotonic relationship is more than a diagnostic: it directly motivates the algorithm developed in the next section and, in particular, its stopping criterion, whereby the procedure terminates once the change in L_2 -norm deviation between successive iterations falls below a threshold ϵ —a mechanism we detail below.

Filtering potential outliers from unlabeled wild data. Building on the monotonic behavior established in Figure 1, we cast outlier identification as the following optimization problem, whose solution yields the outlier subset $\mathcal{S}_{\text{out}}^*$ in $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function and $\hat{y}_{\tilde{\mathbf{x}}_i}$ the predicted label of a wild sample $\tilde{\mathbf{x}}_i$; for notational brevity we write $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$. Intuitively, (4) seeks the subset $\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}$ whose gradient EWM lies closest to the reference gradient $\bar{\nabla}_{\text{in}}$. Given the monotonic relationship of Figure 1, the minimizer $\mathcal{S}_{\text{in}}^*$ is expected to recover the InD portion of $\mathcal{S}_{\text{wild}}$, so that its complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ isolates the OOD portion.

Because exhaustively solving (4) is computationally prohibitive as $|\mathcal{S}_{\text{wild}}|$ grows, we introduce a tractable greedy approximation based on a leave-one-out criterion, given in Algorithm 1. The algorithm performs iterative outlier detection by measuring, at each step, how the removal of individual wild samples reshapes the gradient EWM relative to the InD reference.

At each iteration the algorithm computes the EWM of the current wild-set gradients and compares it against the reference gradient $\bar{\nabla}_{\text{in}}$. Let $d_t =$

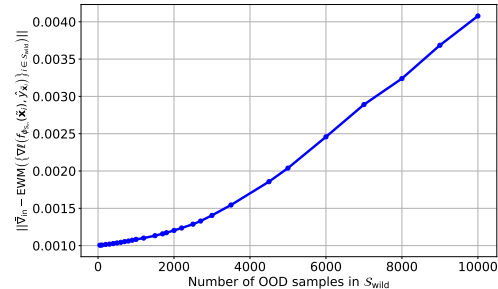


Figure 1: L_2 -norm deviation between the reference InD gradient $\bar{\nabla}_{\text{in}}$ and the EWM of the wild-set gradients increases monotonically as OOD samples are added to $\mathcal{S}_{\text{wild}}$ (CIFAR-10 InD, SVHN OOD, Wide ResNet), motivating both the filtering objective and the stopping criterion.

175 $\|\text{EWM}(G_S) - \bar{\nabla}_{\text{in}}\|$ denote the L_2 distance between
 176 $\bar{\nabla}_{\text{in}}$ and the EWM of the gradients remaining in the wild set $\mathcal{S}$. For each candidate $i \in \mathcal{S}$, the algorithm
 177 evaluates the leave-one-out EWM $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ and the resulting distance $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$,
 178 yielding the induced reduction $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$; the k samples with the largest
 179 reductions are flagged as OOD and removed. This process repeats until no sample yields a significant
 180 reduction δ_i or a maximum iteration count is reached. The convergence criterion requires the inter-
 181 iteration change in L_2 distance to fall below a predefined threshold ϵ . This criterion follows directly
 182 from the monotonic trend observed in Figure 1: once a substantial fraction of OOD samples has been
 183 removed, the L_2 distance settles to a small value, precisely marking the point at which the algorithm
 184 should halt so as to avoid erroneously excising InD samples and thereby degrading performance.

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}$, $G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i}})$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$
 (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_S) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for** each $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\hat{\mathcal{S}}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

185 3.2 Training the OOD Detector with Candidate Outliers

186 Having extracted the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from the wild data, we train the OOD detector g_θ to
 187 maximally separate InD data from candidate outliers, following the protocol of Du et al. (2024a).
 188 The objective enforces separability at the decision boundary (thresholded at 0), driving g_θ toward
 189 positive outputs on labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs on candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$.
 190 Concretely, the loss is

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \tag{5}$$

191 Because the 0/1 loss is non-differentiable, we replace it with a differentiable sigmoid-based binary
 192 surrogate, which preserves alignment with the original objective while rendering the optimization
 193 tractable. The framework simultaneously retains the InD risk of Equation 2, thereby safeguarding
 194 predictive accuracy on InD samples. This unified objective substantially strengthens the generalization
 195 of g_θ and enables robust detection of OOD samples drawn from $\mathbb{P}_{\text{out}}$, as our experiments confirm.

196 4 Theoretical Analysis

197 We now formalize the theoretical guarantees underlying Medix’s filtering stage. The two theorems
 198 below furnish provable upper bounds on the misclassification rates for InD and OOD points respec-
 199 tively, and taken together they establish the two-sided robustness of EWM filtering: neither too many
 200 inliers are discarded nor too many outliers retained. Complete proofs are deferred to Appendix C.

201 **Theorem 4.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points*

in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

202

203 Theorem 4.1 shows that the inlier misclassification rate is governed by exactly two effects: a
 204 *concentration term*, which captures statistical fluctuations in the InD gradients and vanishes as the
 205 sample size grows, and a *contamination term*, which quantifies the influence of OOD points on the
 206 EWM. The bound stays controlled for any contamination ratio $\pi < 0.5$, a decisive property that
 207 underscores the robustness of median-based filtering to substantial OOD presence. This naturally
 208 raises the complementary question: *how many true outliers evade the filter and are mistakenly*
 209 *retained?* The next theorem settles it by bounding the OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

210 Theorem 4.2 complements the preceding result by bounding the fraction of OOD points mistakenly
 211 retained as InD. Together, Theorems 4.1 and 4.2 constitute a two-sided robustness guarantee for
 212 Medix filtering, in which the inlier and outlier misclassification rates are jointly governed by the
 213 balance among three interpretable effects:

214 **Contamination Effect.** In Theorem 4.1, the median-based criterion remains stable even when the
 215 wild dataset contains a large fraction π of OOD points, provided $\pi < 0.5$. This is encoded by the
 216 term $\pi/[2(1 - \pi)]$, which grows with π and captures the risk that OOD gradients skew the EWM.
 217 Theorem 4.2 carries a symmetric penalty $\frac{1-\pi}{2\pi}$, quantifying the mirror-image difficulty of isolating
 218 OOD points when they are underrepresented.

219 **Concentration Effect.** Both bounds exploit the sub-Gaussian structure of the gradient coordinates.
 220 For InD data, this guarantees that the bulk of gradients concentrate near $\bar{\nabla}_{\text{in}}$, keeping the median
 221 stable even at finite sample sizes; for OOD data, concentration around μ_{out} lets us quantify the risk
 222 that a misaligned OOD sample slips past the filter. Both concentration terms decay at the parametric
 223 rate $1/\sqrt{m}$, with m the number of samples from the corresponding class.

224 **Separation Effect.** Unique to Theorem 4.2, the separation effect specifies how far the OOD mean
 225 gradient must lie from the InD mean gradient for OOD samples to be reliably rejected. The exponential
 226 factor $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ encodes this trade-off precisely: the greater the separation between
 227 the two gradient distributions, the smaller the probability that an OOD gradient deceives the filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering achieves robust separation of InD and OOD samples in unlabeled mixtures. The fraction of InD samples misclassified as outliers is bounded by the contamination and concentration effects, while the fraction of OOD points incorrectly retained is bounded by an exponential separation term, a concentration term, and a reverse contamination effect. Together, these results provide rigorous theoretical assurance that Medix simultaneously minimizes both error types under mild assumptions.

228

Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates. The coordinate-wise sub-Gaussianity of InD gradients is the linchpin of our concentration bounds and, more broadly, of robust statistical estimation. This assumption is well supported empirically. Figure 4a shows that the histogram of InD gradient values is bell-shaped and tightly concentrated about the mean, exhibiting the light, exponentially decaying tails that characterize sub-Gaussian random variables and that make extreme gradient values rare. Figure 4b reinforces this via a Q-Q plot (Wilk and Gnanadesikan, 1968) comparing the empirical gradient quantiles against the theoretical Gaussian quantiles: the near-exact alignment with the 45-degree reference line indicates that the empirical distribution closely tracks a Gaussian, and since Gaussians are a subclass of sub-Gaussian variables, the assumption is empirically justified. *Moreover, we establish a looser variant of these bounds that **dispenses with the sub-Gaussian assumption** entirely, requiring only bounded second moments; this appears as Theorem C.3 (Appendix C.3). Although the rates degrade in this weaker regime, the core robustness guarantee of Medix persists, demonstrating that our conclusions extend well beyond the sub-Gaussian setting.*

243 5 Experiments

244 This section establishes the efficacy of Medix across a broad range of InD–OOD dataset pairs,
245 benchmarking it against 20 widely-used baselines. All experiments run on NVIDIA A100-SXM4-
246 80GB GPUs, and the code required to reproduce our results will be released upon acceptance.

247 5.1 Models, Datasets, and Baselines

248 **Datasets.** To ensure a fair comparison with prior work, we adopt the exact experimental protocol of
249 Katz-Samuels et al. (2022a), the study that introduced the problem of learning OOD detectors from
250 wild data. We use CIFAR-10 and CIFAR-100 as InD datasets ($\mathbb{P}_{\text{in}}$), and for OOD testing we employ
251 a diverse suite of natural-image datasets: PLACES365 (Zhou et al., 2017), SVHN, TEXTURES
252 (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) ($\mathbb{P}_{\text{out}}$). To simulate wild
253 data ($\mathbb{P}_{\text{wild}}$), we mix a subset of InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) at a default contamination level
254 $\pi = 0.5$. Concretely, when PLACES365 serves as the OOD test set, we form the wild mixture
255 by combining CIFAR with PLACES365 and evaluate on PLACES365; this procedure is repeated
256 for every OOD dataset and every baseline. Each InD CIFAR set is partitioned into two disjoint
257 halves: the first 25,000 images train $\phi_{\mathcal{S}_{\text{in}}}$, while the remainder form the wild mixture $\mathcal{S}_{\text{wild}}$. Gradients
258 are computed on the penultimate-layer weights, which prior work has shown to be particularly
259 informative for OOD detection (Huang et al., 2021a).

260 **Evaluation metrics.** We report three standard metrics: (1) the False Positive Rate (FPR95↓) of
261 OOD samples at a 95% True Positive Rate on InD samples, (2) the Area Under the Receiver Operating
262 Characteristic Curve (AUROC↑), and (3) InD Classification Accuracy (InD Acc↑). The first two
263 quantify OOD detection quality, while the third verifies that InD performance is preserved.

264 **Baselines.** Our comparison spans a broad and competitive set of baselines, organized by whether
265 they are trained on InD data alone or on both InD and wild data. Among methods trained exclusively
266 on InD data ($\mathbb{P}_{\text{in}}$), we compare Medix against a comprehensive range of established detectors:
267 Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018),
268 Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021),
269 DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all built
270 on a softmax cross-entropy-trained model. For completeness we further include contrastive-loss
271 methods, namely CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). Among methods that leverage
272 both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and
273 energy-regularization learning (Liu et al., 2020b), which regularize training by promoting lower
274 confidence or higher energy on outlier data; against WOODS (Katz-Samuels et al., 2022a), which
275 introduced the notion of wild unlabeled data and exploits it via constrained optimization; and against
276 the more recent CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), which we add for a thorough
277 and up-to-date evaluation.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), we instantiate the InD classifier $\phi_{S_{in}}$ as a Wide ResNet (Zagoruyko, 2016) with 40 layers and width factor 2, trained by stochastic gradient descent with momentum 0.9, weight decay 0.0005, and initial learning rate 0.1 for 100 epochs under cosine learning-rate decay, batch size 128, and dropout rate 0.3. The Medix hyperparameters ϵ and k are selected from $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and $\{4k, 7k, 10k, 20k\}$ respectively, chosen with regard to dataset size and to maximize OOD performance. The OOD classifier g_θ is initialized from the pre-trained InD classifier $\phi_{S_{in}}$ and augmented with a linear layer that performs binary classification on the penultimate-layer features. It is fine-tuned for 100 epochs at learning rate 0.001 according to Equation (5), combining the binary classification loss with the InD classification loss at a relative weight of 10 on the binary term; all remaining training parameters match those used for $\phi_{S_{in}}$.

Table 1: OOD detection performance of Medix versus baselines with CIFAR-10 as InD data, reported per OOD test set (SVHN, PLACES365, LSUN-C, LSUN-Resize, TEXTURES) and as an average, alongside InD accuracy. All numbers are averaged over five runs; the best result in each column is in **bold**, and lower FPR95 / higher AUROC is better. Medix attains the best FPR95 and AUROC on every OOD set.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
	Using P_{in} only												
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
	Using P_{in} and P_{wild}												
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance of Medix versus baselines with CIFAR-100 as InD data, reported per OOD test set and as an average, alongside InD accuracy. All numbers are averaged over five runs; the best result in each column is in **bold**, and lower FPR95 / higher AUROC is better. Medix again achieves the best FPR95 and AUROC across every OOD set, including the strongest wild-data baselines.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using P_{in} and P_{wild}													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

5.3 Results

Our principal results on CIFAR-100 appear in Table 2, where Medix outperforms every OOD detection baseline by a decisive margin. Two patterns stand out. First, methods that exploit both InD and wild data substantially surpass those confined to InD data: relative to the InD-only baseline KNN+, Medix lowers FPR95 by 52.31% on PLACES365 and by 38.24% on TEXTURES, quantifying the value of in-the-wild data as a regularizer. Second, Medix also dominates the strongest wild-data competitors: it attains an average FPR95 of 5.42% on CIFAR-100, a 1.32% absolute improvement over WOODS, while sustaining a competitive InD accuracy of 73.33%. The modest accuracy gap is

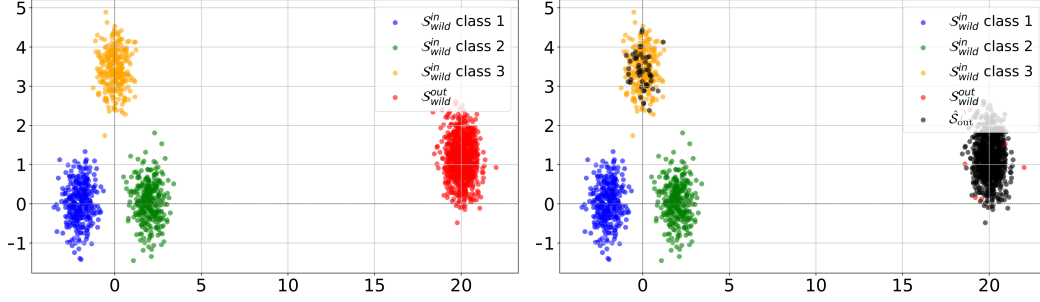


Figure 2: Medix applied to a synthetic 2D wild mixture. (a) The wild set comprises InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$, drawn from three multivariate Gaussians, together with OOD data $\mathcal{S}_{\text{wild}}^{\text{out}}$. (b) The outliers $\hat{\mathcal{S}}_{\text{out}}$ flagged by Medix (black) align closely with the ground-truth OOD points: only 12.5% of $\hat{\mathcal{S}}_{\text{out}}$ are in fact InD points, illustrating the low extraction error predicted by our theory.

fully accounted for by the experimental protocol—Medix is trained on 25,000 labeled InD samples, whereas the InD-only baselines use the full 50,000-sample CIFAR-100 training set—and is therefore a consequence of the wild-data setting rather than a limitation of the method. The CIFAR-10 results in Table 1 tell the same story: Medix surpasses all baselines, improving FPR95 over WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES, further confirming its effectiveness at detecting OOD samples.

A representative visual example of Medix. To make the behavior of Algorithm 1 (Outlier Extraction) directly interpretable, we construct a deliberately simple two-dimensional synthetic experiment. The InD data is drawn from three multivariate Gaussians—one per class—with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$ and shared covariance $0.25 \cdot I$, while the OOD data follows $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. As Figure 2 shows, Medix recovers the outliers almost perfectly: it correctly flags 87.5% of the true OOD samples, with the extracted set containing only 12.5% InD contamination. This concrete instance vividly substantiates the low extraction error established by our theoretical analysis.

Additional studies and insights. Owing to space constraints, we defer a substantial body of further experiments to Appendix A, each of which reinforces the conclusions above: (1) hyperparameter-selection ablations and a sensitivity analysis (Appendix A.2) establishing Medix’s strong robustness to its hyperparameters; (2) a comparison of EWM against the geometric median (Appendix A.1), showing that EWM is more sensitive to distributional shift and hence the more effective and reliable filtering choice; (3) a comprehensive comparison against methods trained with competitive contrastive objectives (Appendix A.3), in which Medix outperforms even these strong baselines by a significant margin; (4) an evaluation on large-scale data under the demanding unseen-OOD setting $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ (Appendix A.4), where Medix again leads by a wide margin; (5) an assessment of Medix’s computation and memory efficiency (Appendix A.6); (6) an analysis of the impact of pseudo-label quality (Appendix A.5), demonstrating resilience to noisy or low-confidence labels; and (7) a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

OOD detection has attracted intense and sustained interest in recent years (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djurisić et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One prominent family derives scoring functions to gauge the distributional origin of an input, spanning distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Sehwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boulton, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

A second family imposes regularization during training (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a), for instance by suppressing model confidence (Lee et al., 2017; Hendrycks et al., 2019) or raising its energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data. Critically, most of these methods presuppose access to a clean auxiliary OOD dataset—an assumption Medix does not require.

A related line of work relaxes the labeled-only assumption by incorporating unlabeled wild data (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023; Katz-Samuels et al., 2022a; Geng et al., 2025), yet these studies stop short of an explicit outlier-detection mechanism. Du et al. (2024a,b) do introduce explicit outlier filtering, but their thresholding technique differs fundamentally from ours: we detect outliers through a new median-centric criterion. A further distinction is structural. Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, in which each batch contains a fixed ratio of InD and OOD samples; with large outsourced datasets such structure is unavailable, since data is mixed randomly across the corpus. Medix removes this requirement, operating directly under dataset-level mixing. Finally, many works adopt positive-unlabeled learning, training classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022); the essential difference is that these methods only separate $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ and do not simultaneously train an OOD classifier. In contrast, Medix couples a median-centric outlier-identification procedure—with provably low error rates—to the training of a dedicated OOD detector.

7 Conclusions

We introduced Medix, a median-centric framework for OOD detection that turns unlabeled in-the-wild data into a reliable training signal. By exploiting the inherent robustness of the median operation, Medix filters outliers from mixed unlabeled data and thereby enables the training of a dependable OOD detector. Our theoretical analysis furnished provable bounds on the inlier misclassification rate, showing that Medix remains robust even under substantial OOD contamination—up to 50%—with errors controlled by sub-Gaussian concentration and contamination effects, and complemented these with matching bounds on the OOD misclassification rate that guarantee accurate isolation of outliers under clear separation. Empirically, Medix surpassed all 20 baselines across diverse benchmarks: it reduced average FPR95 by 40.98% relative to strong baselines such as KNN+, outperformed state-of-the-art wild-data methods including WOODS and DRL, and attained an outlier-extraction error rate as low as 12.5%. Together, these theoretical and empirical results position median-centric filtering as a principled and effective foundation for OOD detection from unlabeled data.

References

- Abbas, M., Azmat, M., Horesh, R., and Yurochkin, M. (2025). Out-of-distribution detection using synthetic data generation. In *Second Conference on Language Modeling*.
- Acharya, A., Dhillon, I. S., and Sanghavi, S. (2024). Geometric median (gm) matching for robust data pruning. *arXiv preprint arXiv:2406.17188*.
- Acharya, A., Sanghavi, S., Jing, L., Bhushanam, B., Choudhary, D., Rabbat, M., and Dhillon, I. (2022). Positive unlabeled contrastive learning. *arXiv preprint arXiv:2206.01206*.
- Bai, H., Canal, G., Du, X., Kwon, J., Nowak, R. D., and Li, Y. (2023). Feed two birds with one scone: Exploiting wild data for both out-of-distribution generalization and detection. In *International Conference on Machine Learning*, pages 1454–1471. PMLR.
- Bendale, A. and Boulton, T. E. (2016). Towards open set deep networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1563–1572.
- Chapel, L., Alaya, M. Z., and Gasso, G. (2020). Partial optimal transport with applications on positive-unlabeled learning. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*, volume 33, pages 2903–2913. Curran Associates, Inc.
- Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., and Vedaldi, A. (2014). Describing textures in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3606–3613.

386 Djuricic, A., Bozanic, N., Ashok, A., and Liu, R. (2022). Extremely simple activation shaping for out-of-
387 distribution detection. *arXiv preprint arXiv:2209.09858*.

388 Djuricic, A., Bozanic, N., Ashok, A., and Liu, R. (2023). Extremely simple activation shaping for out-of-
389 distribution detection. In *The Eleventh International Conference on Learning Representations*.

390 Du, X., Fang, Z., Diakonikolas, I., and Li, Y. (2024a). How does unlabeled data provably help out-of-distribution
391 detection? In *The Twelfth International Conference on Learning Representations*.

392 Du, X., Ghosh, R., Sim, R., Salem, A., Carvalho, V., Lawton, E., Li, Y., and Stokes, J. W. (2024b). Vlmguard:
393 Defending vlms against malicious prompts via unlabeled data. *arXiv preprint arXiv:2410.00296*.

394 Du, X., Gozum, G., Ming, Y., and Li, Y. (2022a). Siren: Shaping representations for detecting out-of-distribution
395 objects. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in
396 Neural Information Processing Systems*, volume 35, pages 20434–20449. Curran Associates, Inc.

397 Du, X., Sun, Y., Zhu, J., and Li, Y. (2023). Dream the impossible: Outlier imagination with diffusion models.
398 In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S., editors, *Advances in Neural
399 Information Processing Systems*, volume 36, pages 60878–60901. Curran Associates, Inc.

400 Du, X., Wang, X., Gozum, G., and Li, Y. (2022b). Unknown-aware object detection: Learning what you don’t
401 know from videos in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern
402 Recognition*, pages 13678–13688.

403 Du, X., Wang, Z., Cai, M., and Li, S. (2022c). Towards unknown-aware learning with virtual outlier synthesis.
404 In *International Conference on Learning Representations*.

405 Fan, Y., Kukleva, A., Dai, D., and Schiele, B. (2023). Ssb: Simple but strong baseline for boosting performance
406 of open-set semi-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer
407 Vision (ICCV)*, pages 16068–16078.

408 Fang, Z., Li, Y., Lu, J., Dong, J., Han, B., and Liu, F. (2022). Is out-of-distribution detection learnable? In
409 Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural
410 Information Processing Systems*, volume 35, pages 37199–37213. Curran Associates, Inc.

411 Fort, S., Ren, J., and Lakshminarayanan, B. (2021). Exploring the limits of out-of-distribution detection.
412 *Advances in Neural Information Processing Systems*, 34:7068–7081.

413 Gal, Y. and Ghahramani, Z. (2016). Dropout as a bayesian approximation: Representing model uncertainty
414 in deep learning. In Balcan, M. F. and Weinberger, K. Q., editors, *Proceedings of The 33rd International
415 Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–
416 1059, New York, New York, USA. PMLR.

417 Galil, I., Dabbah, M., and El-Yaniv, R. (2023). A framework for benchmarking class-out-of-distribution detection
418 and its application to imagenet. In *The Eleventh International Conference on Learning Representations*.

419 Garg, S., Balakrishnan, S., and Lipton, Z. (2022). Domain adaptation under open set label shift. In Koyejo,
420 S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information
421 Processing Systems*, volume 35, pages 22531–22546. Curran Associates, Inc.

422 Garg, S., Wu, Y., Smola, A. J., Balakrishnan, S., and Lipton, Z. (2021). Mixture proportion estimation and
423 pu learning: a modern approach. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan,
424 J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 8532–8544. Curran
425 Associates, Inc.

426 Geifman, Y. and El-Yaniv, R. (2019). SelectiveNet: A deep neural network with an integrated reject option. In
427 Chaudhuri, K. and Salakhutdinov, R., editors, *Proceedings of the 36th International Conference on Machine
428 Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2151–2159. PMLR.

429 Geng, C., Li, Q., Wang, X., Liang, D., Chen, S., and Yuen, P. C. (2025). Lod: Loss-difference ood detection by
430 intentionally label-noisifying unlabeled wild data. *arXiv preprint arXiv:2505.12952*.

431 Ghosal, S. S., Sun, Y., and Li, Y. (2024). How to overcome curse-of-dimensionality for out-of-distribution
432 detection? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19849–19857.

433 Gong, T., Wang, G., Ye, J., Xu, Z., and Lin, M. (2018). Margin based pu learning. In *Proceedings of the AAAI
434 Conference on Artificial Intelligence*, volume 32.

- 435 Hang, J.-Y. and Zhang, M.-L. (2024). Binary decomposition: A problem transformation perspective for open-set
436 semi-supervised learning. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and
437 Berkenkamp, F., editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235
438 of *Proceedings of Machine Learning Research*, pages 17505–17518. PMLR.
- 439 He, R., Li, R., Han, Z., Yang, X., and Yin, Y. (2023). Topological structure learning for weakly-supervised
440 out-of-distribution detection. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages
441 4858–4866.
- 442 Hein, M., Andriushchenko, M., and Bitterwolf, J. (2019). Why relu networks yield high-confidence predictions
443 far away from the training data and how to mitigate the problem. In *Proceedings of the IEEE/CVF conference*
444 *on computer vision and pattern recognition*, pages 41–50.
- 445 Hendrycks, D. and Gimpel, K. (2016). A baseline for detecting misclassified and out-of-distribution examples in
446 neural networks. *arXiv preprint arXiv:1610.02136*.
- 447 Hendrycks, D. and Gimpel, K. (2017). A baseline for detecting misclassified and out-of-distribution examples in
448 neural networks. In *International Conference on Learning Representations*.
- 449 Hendrycks, D., Mazeika, M., and Dietterich, T. (2019). Deep anomaly detection with outlier exposure. In
450 *International Conference on Learning Representations*.
- 451 Hsieh, C.-J., Natarajan, N., and Dhillon, I. (2015). Pu learning for matrix completion. In Bach, F. and Blei, D.,
452 editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings*
453 *of Machine Learning Research*, pages 2445–2453, Lille, France. PMLR.
- 454 Huang, R., Geng, A., and Li, Y. (2021a). On the importance of gradients for detecting distributional shifts in the
455 wild. *Advances in Neural Information Processing Systems*, 34:677–689.
- 456 Huang, R., Geng, A., and Li, Y. (2021b). On the importance of gradients for detecting distributional shifts in the
457 wild. In Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information*
458 *Processing Systems*.
- 459 Huber, P. J. (1964). Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–
460 101.
- 461 Jeong, T. and Kim, H. (2020). Ood-maml: Meta-learning for few-shot out-of-distribution detection and
462 classification. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in*
463 *Neural Information Processing Systems*, volume 33, pages 3907–3916. Curran Associates, Inc.
- 464 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022a). Training ood detectors in their natural habitats.
465 In *International Conference on Machine Learning*, pages 10848–10865. PMLR.
- 466 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022b). Training OOD detectors in their natural
467 habitats. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings*
468 *of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning*
469 *Research*, pages 10848–10865. PMLR.
- 470 Kim, S., Kim, D., Cho, M., and Kwak, S. (2020). Proxy anchor loss for deep metric learning. In *2020 IEEE/CVF*
471 *Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3235–3244.
- 472 Kristiadi, A., Hein, M., and Hennig, P. (2020). Being bayesian, even just a bit, fixes overconfidence in relu
473 networks. In *International conference on machine learning*, pages 5436–5446. PMLR.
- 474 Krizhevsky, A., Hinton, G., et al. (2009). Learning multiple layers of features from tiny images.
- 475 Lakshminarayanan, B., Pritzel, A., and Blundell, C. (2017). Simple and scalable predictive uncertainty estimation
476 using deep ensembles. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S.,
477 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates,
478 Inc.
- 479 Lee, K., Lee, H., Lee, K., and Shin, J. (2017). Training confidence-calibrated classifiers for detecting out-of-
480 distribution samples. *arXiv preprint arXiv:1711.09325*.
- 481 Lee, K., Lee, K., Lee, H., and Shin, J. (2018a). A simple unified framework for detecting out-of-distribution
482 samples and adversarial attacks. *Advances in neural information processing systems*, 31.

483 Lee, K., Lee, K., Lee, H., and Shin, J. (2018b). A simple unified framework for detecting out-of-distribution
484 samples and adversarial attacks. In Bengio, S., Wallach, H., Larochelle, H., Grauman, K., Cesa-Bianchi, N.,
485 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates,
486 Inc.

487 Letouzey, F., Denis, F., and Gilleron, R. (2000). Learning from positive and unlabeled examples. In *International*
488 *Conference on Algorithmic Learning Theory*, pages 71–85. Springer.

489 Liang, S., Li, Y., and Srikant, R. (2018). Enhancing the reliability of out-of-distribution image detection in
490 neural networks. In *International Conference on Learning Representations*.

491 Liu, J., Lin, Z., Padhy, S., Tran, D., Bedrax Weiss, T., and Lakshminarayanan, B. (2020a). Simple and principled
492 uncertainty estimation with deterministic deep learning via distance awareness. In Larochelle, H., Ranzato,
493 M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*,
494 volume 33, pages 7498–7512. Curran Associates, Inc.

495 Liu, W., Wang, X., Owens, J., and Li, Y. (2020b). Energy-based out-of-distribution detection. *Advances in*
496 *neural information processing systems*, 33:21464–21475.

497 Liu, W., Wang, X., Owens, J., and Li, Y. (2020c). Energy-based out-of-distribution detection. In Larochelle,
498 H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing*
499 *Systems*, volume 33, pages 21464–21475. Curran Associates, Inc.

500 Maddox, W. J., Izmailov, P., Garipov, T., Vetrov, D. P., and Wilson, A. G. (2019). A simple baseline for bayesian
501 uncertainty in deep learning. In Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., and
502 Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

503 Malinin, A. and Gales, M. (2018). Predictive uncertainty estimation via prior networks. In Bengio, S., Wallach,
504 H., Larochelle, H., Grauman, K., Cesa-Bianchi, N., and Garnett, R., editors, *Advances in Neural Information*
505 *Processing Systems*, volume 31. Curran Associates, Inc.

506 Malinin, A. and Gales, M. (2019). Reverse kl-divergence training of prior networks: Improved uncertainty and
507 adversarial robustness. In Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., and Garnett,
508 R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

509 Meinke, A. and Hein, M. (2020). Towards neural networks that provably know when they don't know. In
510 *International Conference on Learning Representations*.

511 Ming, Y., Fan, Y., and Li, Y. (2022). POEM: Out-of-distribution detection with posterior sampling. In
512 Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th*
513 *International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*,
514 pages 15650–15665. PMLR.

515 Ming, Y. and Li, Y. (2024). How does fine-tuning impact out-of-distribution detection for vision-language
516 models? *International Journal of Computer Vision*, 132(2):596–609.

517 Ming, Y., Sun, Y., Dia, O., and Li, Y. (2023). How to exploit hyperspherical embeddings for out-of-distribution
518 detection? In *The Eleventh International Conference on Learning Representations*.

519 Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. (2011). Reading digits in natural images
520 with unsupervised feature learning. In *NIPS workshop on deep learning and unsupervised feature learning*,
521 volume 2011, page 4. Granada.

522 Nguyen, A., Yosinski, J., and Clune, J. (2015). Deep neural networks are easily fooled: High confidence
523 predictions for unrecognizable images. In *Proceedings of the IEEE conference on computer vision and pattern*
524 *recognition*, pages 427–436.

525 Niu, G., Du Plessis, M. C., Sakai, T., Ma, Y., and Sugiyama, M. (2016). Theoretical comparisons of positive-
526 unlabeled learning against positive-negative learning. *Advances in neural information processing systems*,
527 29.

528 Peng, B., Luo, Y., Zhang, Y., Li, Y., and Fang, Z. (2024). Conjnorm: Tractable density estimation for out-of-
529 distribution detection. In *The Twelfth International Conference on Learning Representations*.

530 Plessis, M. D., Niu, G., and Sugiyama, M. (2015). Convex formulation for learning from positive and unlabeled
531 data. In Bach, F. and Blei, D., editors, *Proceedings of the 32nd International Conference on Machine Learning*,
532 volume 37 of *Proceedings of Machine Learning Research*, pages 1386–1394, Lille, France. PMLR.

533 Ren, J., Fort, S., Liu, J., Roy, A. G., Padhy, S., and Lakshminarayanan, B. (2021). A simple fix to mahalanobis
534 distance for improving near-ood detection. *arXiv preprint arXiv:2106.09022*.

535 Ren, J., Luo, J., Zhao, Y., Krishna, K., Saleh, M., Lakshminarayanan, B., and Liu, P. J. (2022). Out-of-distribution
536 detection and selective generation for conditional language models. In *The Eleventh International Conference*
537 *on Learning Representations*.

538 Saito, K., Kim, D., and Saenko, K. (2021). Openmatch: Open-set semi-supervised learning with open-set
539 consistency regularization. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W.,
540 editors, *Advances in Neural Information Processing Systems*, volume 34, pages 25956–25967. Curran
541 Associates, Inc.

542 Sehwag, V., Chiang, M., and Mittal, P. (2021). {SSD}: A unified framework for self-supervised outlier detection.
543 In *International Conference on Learning Representations*.

544 Sun, Y., Guo, C., and Li, Y. (2021). React: Out-of-distribution detection with rectified activations. *Advances in*
545 *Neural Information Processing Systems*, 34:144–157.

546 Sun, Y. and Li, Y. (2022). Dice: Leveraging sparsification for out-of-distribution detection. In *European*
547 *Conference on Computer Vision*, pages 691–708. Springer.

548 Sun, Y., Ming, Y., Zhu, X., and Li, Y. (2022). Out-of-distribution detection with deep nearest neighbors. In
549 *International Conference on Machine Learning*, pages 20827–20840. PMLR.

550 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020a). Csi: Novelty detection via contrastive learning on distributionally
551 shifted instances. *Advances in neural information processing systems*, 33:11839–11852.

552 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020b). Csi: Novelty detection via contrastive learning on distributionally
553 shifted instances. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in*
554 *Neural Information Processing Systems*, volume 33, pages 11839–11852. Curran Associates, Inc.

555 Tao, L., Du, X., Zhu, J., and Li, Y. (2023). Non-parametric outlier synthesis. In *The Eleventh International*
556 *Conference on Learning Representations*.

557 Uppaal, R., Hu, J., and Li, Y. (2023). Is fine-tuning needed? pre-trained language models are near perfect
558 for out-of-domain detection. In Rogers, A., Boyd-Graber, J., and Okazaki, N., editors, *Proceedings of the*
559 *61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages
560 12813–12832, Toronto, Canada. Association for Computational Linguistics.

561 Van Amersfoort, J., Smith, L., Teh, Y. W., and Gal, Y. (2020). Uncertainty estimation using a single deep
562 deterministic neural network. In *International conference on machine learning*, pages 9690–9700. PMLR.

563 Wang, H., Li, Z., Feng, L., and Zhang, W. (2022a). Vim: Out-of-distribution with virtual-logit matching. In
564 *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4921–4930.

565 Wang, H., Liu, W., Bocchieri, A., and Li, Y. (2021). Can multi-label classification networks know what they
566 don’t know? *Advances in Neural Information Processing Systems*, 34:29074–29087.

567 Wang, Q., Fang, Z., Zhang, Y., Liu, F., Li, Y., and Han, B. (2023a). Learning to augment distributions for
568 out-of-distribution detection. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine,
569 S., editors, *Advances in Neural Information Processing Systems*, volume 36, pages 73274–73286. Curran
570 Associates, Inc.

571 Wang, Q., Liu, F., Zhang, Y., Zhang, J., Gong, C., Liu, T., and Han, B. (2022b). Watermarking for out-of-
572 distribution detection. *Advances in Neural Information Processing Systems*, 35:15545–15557.

573 Wang, Q., Ye, J., Liu, F., Dai, Q., Kalander, M., Liu, T., HAO, J., and Han, B. (2023b). Out-of-distribution
574 detection with implicit outlier transformation. In *The Eleventh International Conference on Learning*
575 *Representations*.

576 Wang, Y., Zou, J., Lin, J., Ling, Q., Pan, Y., Yao, T., and Mei, T. (2022c). Out-of-distribution detection via
577 conditional kernel independence model. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K.,
578 and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages 36411–36425.
579 Curran Associates, Inc.

580 Wang, Z., Xiang, L., Huang, L., Mao, J., Xiao, L., and Yamasaki, T. (2024). Scomatch: Alleviating overtrusting
581 in open-set semi-supervised learning. In *European Conference on Computer Vision*, pages 217–233. Springer.

Wei, H., Xie, R., Cheng, H., Feng, L., An, B., and Li, Y. (2022). Mitigating neural network overconfidence with logit normalization. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 23631–23644. PMLR.

Wen, Y., Tran, D., and Ba, J. (2020). Batchensemble: an alternative approach to efficient ensemble and lifelong learning. In *International Conference on Learning Representations*.

Wilk, M. B. and Gnanadesikan, R. (1968). Probability plotting methods for the analysis of data. *Biometrika*, 55(1):1–17.

Wu, Q., Chen, Y., Yang, C., and Yan, J. (2023). Energy-based out-of-distribution detection for graph neural networks. In *The Eleventh International Conference on Learning Representations*.

Xu, D. and Denil, M. (2021). Positive-unlabeled reward learning. In *Conference on Robot Learning*, pages 205–219. PMLR.

Yang, J., Wang, H., Feng, L., Yan, X., Zheng, H., Zhang, W., and Liu, Z. (2021). Semantically coherent out-of-distribution detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8301–8309.

Yang, J., Wang, P., Zou, D., Zhou, Z., Ding, K., Peng, W., Wang, H., Chen, G., Li, B., Sun, Y., et al. (2022). Openood: Benchmarking generalized out-of-distribution detection. *Advances in Neural Information Processing Systems*, 35:32598–32611.

Yang, J., Zhou, K., Li, Y., and Liu, Z. (2024). Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 132(12):5635–5662.

Yu, F., Seff, A., Zhang, Y., Song, S., Funkhouser, T., and Xiao, J. (2015). Lsun: Construction of a large-scale image dataset using deep learning with humans in the loop. *arXiv preprint arXiv:1506.03365*.

Zagoruyko, S. (2016). Wide residual networks. *arXiv preprint arXiv:1605.07146*.

Zhang, J., Yang, J., Wang, P., Wang, H., Lin, Y., Zhang, H., Sun, Y., Du, X., Zhou, K., Zhang, W., et al. (2023). Openood v1. 5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*.

Zhang, Y., Lu, J., Peng, B., Fang, Z., and ming Cheung, Y. (2024). Learning to shape in-distribution feature space for out-of-distribution detection. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.

Zhao, Y., Xu, Q., Jiang, Y., Wen, P., and Huang, Q. (2022). Dist-pu: Positive-unlabeled learning from a label distribution perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14461–14470.

Zheng, H., Wang, Q., Fang, Z., Xia, X., Liu, F., Liu, T., and Han, B. (2023). Out-of-distribution detection learning with unreliable out-of-distribution sources. *Advances in Neural Information Processing Systems*, 36:72110–72123.

Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., and Torralba, A. (2017). Places: A 10 million image database for scene recognition. *IEEE transactions on pattern analysis and machine intelligence*, 40(6):1452–1464.

Zhou, Z., Guo, L.-Z., Cheng, Z., Li, Y.-F., and Pu, S. (2021). Step: Out-of-distribution detection in the presence of limited in-distribution labeled data. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 29168–29180. Curran Associates, Inc.

Zhu, J., Geng, Y., Yao, J., Liu, T., Niu, G., Sugiyama, M., and Han, B. (2023). Diversified outlier exposure for out-of-distribution detection via informative extrapolation. *Advances in Neural Information Processing Systems*, 36:22702–22734.

Zhu, Y., Chen, Y., Xie, C., Li, X., Zhang, R., Xue, H., Tian, X., zheng, b., and Chen, Y. (2022). Boosting out-of-distribution detection with typical features. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages 20758–20769. Curran Associates, Inc.

629 A Additional Studies

630 A.1 Element-wise Median vs Geometric Median

631 A natural question is why Medix employs the element-wise median rather
 632 than the geometric median (Acharya et al., 2024) for outlier detection.
 633 To answer it, we ran a controlled experiment with CIFAR-
 634 10 as the InD dataset and SVHN as the OOD dataset. Here
 635 $\mathcal{S}_{\text{wild}}$ consists of 5k CIFAR-10 samples held disjoint from
 636 the data used to train $\phi_{\mathcal{S}_{\text{in}}}$ (and hence to compute $\bar{\nabla}_{\text{in}}$);
 637 we then incrementally add SVHN OOD samples while
 638 tracking both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the
 639 element-wise median of the wild-set gradients and the
 640 proportion of OOD samples removed. As Figure 3 shows,
 641 at a matched OOD count the element-wise median flags a
 642 substantially higher proportion of OOD samples as outliers
 643 than the geometric median. This confirms that the element-
 644 wise median is markedly more sensitive to distributional
 645 shift, making it the more effective and reliable filtering
 646 criterion for Medix.

647 A.2 Hyperparameter 648 Selection and Sensitivity Analysis

649 We conducted an ablation to quantify Medix’s sensitivity
 650 to the hyperparameters ϵ and k , using CIFAR-100 as the
 651 InD dataset and SVHN as the OOD dataset. As Table 3
 652 shows, Medix is strongly robust to variation in both ϵ and k . It attains a remarkably low FPR95 of
 653 0.16 at $\epsilon = 0.005$, $k = 20000$, and—crucially—perturbing these settings degrades performance only
 654 gracefully rather than sharply: reducing ϵ to 0.0005 or halving k to 10000 yields FPR95 values of
 655 0.20 and 0.68 respectively, each of which still comfortably surpasses every baseline. Thus, while
 656 careful tuning is beneficial, Medix retains its advantage across a wide range of hyperparameter
 657 choices. This insensitivity to exact tuning is a practical asset, making Medix a dependable choice for
 658 real-world deployments in which exhaustive tuning is infeasible.

Table 3: Effect of the hyperparameters ϵ and k on Medix OOD detection (CIFAR-100 InD, SVHN OOD, FPR95 $\downarrow$), with representative InD-only and wild-data baselines for reference. Medix remains far below every baseline across all tested (ϵ, k) settings.

Method	FPR95 $\downarrow$	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

659 A.3 Comprehensive comparison with recent competitive methods

660 We extend our comparison to competitive baselines built on diverse contrastive objectives—SSD+
 661 (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023)—which we
 662 include for completeness despite Medix not employing contrastive learning. We further add the more
 663 recent CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du
 664 et al., 2022c) for a thorough evaluation. For this study we use CIFAR-100 as the InD dataset and train

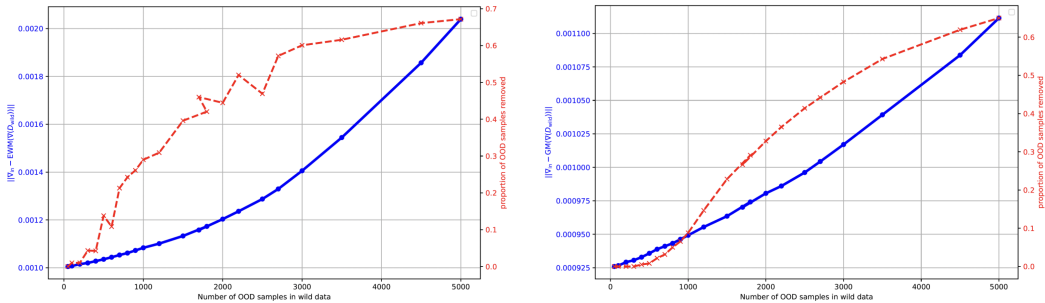


Figure 3: Element-wise median (EWM) versus geometric median (GM) as OOD samples are added to $\mathcal{S}_{\text{wild}}$ (CIFAR-10 InD, SVHN OOD, Wide ResNet): for a matched OOD count, EWM flags a substantially higher proportion of true OOD samples as outliers, evidencing its greater sensitivity to distributional shift.

Table 4: OOD detection performance of Medix versus recent competitive baselines with CIFAR-100 as InD data (ResNet-34), reported per OOD test set and as an average. All numbers are averaged over five runs; the best result in each column is in **bold**, and lower FPR95 / higher AUROC is better. Medix leads on every OOD set, including against methods trained with contrastive objectives.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

the ResNet-34 model¹ following Ming et al. (2023); Zhang et al. (2024): stochastic gradient descent with momentum 0.9 and weight decay 10^{-4} for 500 epochs, initial learning rate 0.5 under cosine scheduling, and batch size 512, reusing the checkpoints of Ming et al. (2023)². As Table 4 shows, Medix outperforms every one of these competitive baselines by a significant margin—including those trained with contrastive objectives—achieving an average FPR95 of 8.68% and an average AUROC of 97.76%. Beyond raw performance, Medix additionally furnishes a provably low error rate, offering theoretical guarantees that most of these alternatives lack.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We now examine whether Medix scales to large-scale wild OOD data under the demanding unseen-OOD setting. This regime is challenging on two counts: (1) it draws on a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), which sharply increases both data complexity and computational demand; and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is deliberately made distributionally distinct from the training-time wild OOD data, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, faithfully mirroring realistic deployment in which the OOD encountered at test time need not match what was seen during training. We take CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and SVHN as the test OOD data. The substantial distributional gap between the large-scale wild OOD data (300K Random Images) and the OOD test data (SVHN) makes this a stringent test of generalization to unseen outliers. Comparing against baselines that likewise leverage wild data during training, Table 5 shows that Medix is clearly superior, delivering a markedly lower FPR95 (41.29 ± 1.2) and a higher AUROC (87.25 ± 0.6)—compelling evidence of its effectiveness at distinguishing InD from OOD data even when the test-time OOD distribution is unseen.

Table 5: OOD detection performance on large-scale unseen OOD data (CIFAR-100 InD, 300K Random Images as wild OOD, SVHN as test OOD), compared against wild-data baselines. Medix roughly halves the FPR of the best baseline while achieving the highest AUROC.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

²<https://github.com>

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

A further question concerns the role of pseudo-label quality $\hat{y}_{\tilde{x}_i}$: does the softmax confidence of the pseudo-labels materially affect OOD filtering, and could a simple pre-filtering step that discards low-confidence pseudo-labels improve robustness? To settle this, we evaluate the effect of removing low-confidence pseudo-labels, using CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. We compute the softmax probability of each pseudo-labeled sample and discard predictions below a confidence threshold of 0.6; this removes 15.98% of the samples from the training set.

Performance is essentially unchanged by this intervention: both variants yield an FPR95 of 0.01%, and filtering nudges AUROC only marginally, from 99.98% to 99.99%. Removing low-confidence pseudo-labels therefore has no meaningful effect on either robustness or OOD detection, demonstrating that Medix is resilient to noisy or low-confidence labels and that additional filtering is unlikely to yield appreciable gains.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We next address whether the Medix filtering phase is computationally feasible and memory-efficient at scale. To this end we profiled the procedure on an NVIDIA A100-SXM4-80GB GPU with roughly 15,000 unlabeled samples per InD–OOD pair, using CIFAR-10 and CIFAR-100 as the InD datasets and LSUN-Resize as the OOD dataset. As Table 6 reports, GPU memory usage stays modest and the filtering phase completes within a tractable wall-clock budget even for thousands of unlabeled samples. Medix filtering is thus computationally practical and imposes no excessive memory overhead, confirming its suitability for large datasets.

Table 6: Profiling of the Medix filtering phase on an NVIDIA A100 80GB GPU ($\approx 15,000$ unlabeled samples per pair): wall-clock time and peak/current GPU memory for two InD–OOD pairs. Memory overhead remains under 100 MB throughout.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix departs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) along three principal axes—objective, treatment of outliers, and technique. First, the objectives differ fundamentally. In OpenMatch Saito et al. (2021), the goal is to cope with outliers while training a standard classifier under semi-supervised learning (SSL); in our setting, the goal is to detect OOD samples in unlabeled wild data and train a dedicated OOD detector. Where SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) seek accurate classification despite outliers, Medix targets OOD detection itself in an open-world regime devoid of labeled OOD data.

Second, the two paradigms treat outliers oppositely. SSL methods train a classifier robust to OOD samples by treating outliers as noise to be suppressed, thereby improving SSL performance; Medix instead explicitly detects OOD samples and actively exploits them to construct a more effective and reliable OOD detection system.

Third, the techniques diverge. OpenMatch rests on consistency regularization—enforcing agreement across stochastic transformations of the same input, formalized as soft constraints on the classifier’s outputs that encourage smooth decision boundaries. Medix instead performs median-based filtering, exploiting the statistical robustness of the median and its insensitivity to extreme values to identify candidate OOD samples, and its accompanying analysis derives error bounds governed by the contamination effect (the OOD proportion) and the concentration effect (the sub-Gaussian behavior of InD gradients), guaranteeing that OOD detection error remains low.

B Broader Impacts and Limitations

As machine learning systems proliferate, robust OOD detection has become indispensable to reliable model performance in real-world applications. Our median-centric framework advances this goal, strengthening OOD handling and, with it, model safety. The significance of this contribution extends beyond its theoretical guarantees to practical deployment across domains such as healthcare, autonomous systems, and finance: by improving OOD detection, Medix addresses a central obstacle to trustworthy deployment and thereby fosters broader confidence in and adoption of machine learning technologies. Promising avenues for future work include coupling Medix with generative models for outlier synthesis and adapting it to dynamic environments in which the OOD distribution drifts over time.

A potential limitation of Medix is its reliance on a mixture of unlabeled InD and OOD data that may, in practice, be noisy, corrupted, or inconsistently sampled. This assumption is not peculiar to our method; it reflects the intrinsic uncertainty of real-world deployment and is a standard starting point in robust learning theory. Critically, we render our assumptions explicit and verifiable—formalizing them in the main theorems and stating a precise geometric separation condition in Remark 4.3—and we relax them substantially in Theorem C.3, which dispenses with sub-Gaussian tails and establishes the resilience of our framework under weaker distributional assumptions. Taken together, these results show that the simplifying assumptions of Medix are both interpretable and relaxable, constituting a principled foundation for future advances in OOD detection under realistic noise.

754 C Proof of Theorem

755 For clarity and consistency across the theoretical and empirical sections, Table 7 consolidates the key
 756 notation used throughout the paper, covering gradient-related quantities, sample sizes, filtering terms,
 757 and the parameters that enter our concentration bounds. Unless stated otherwise, we denote vectors
 758 in boldface, write $\mathcal{X}$ for the input space, and compute all gradients with respect to a fixed pretrained
 759 model f_ϕ .

Table 7: Notation summary. We use bold symbols for vectors and $\mathcal{X}$ to denote the input space.

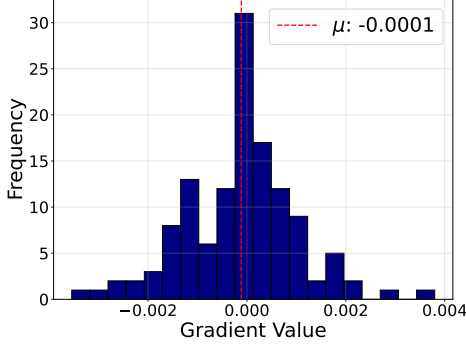
Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

760 C.1 Bound on the Inlier Misclassification Rate

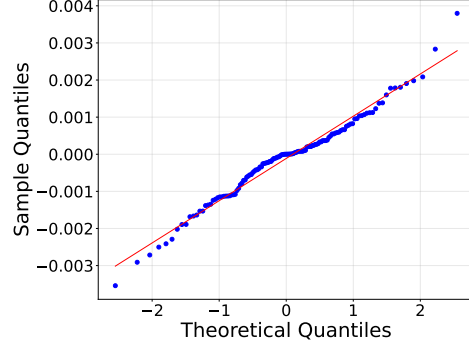
761 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 762 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 763 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 764 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

765 *Proof.* The argument proceeds in two steps. In **Step 1** we establish a high-probability concentration
 766 bound for the gradients of InD samples about their mean $\bar{\nabla}_{\text{in}}$, terming an InD sample whose gradient
 767 falls within this high-probability region a *good inlier*. In **Step 2** we invoke a “swapping” argument
 768 on the EWM objective to show that excluding more than a $(2\eta + \frac{\pi}{2}(1 - \pi))$ fraction of good
 769 inliers—where η denotes the fraction of InD samples with gradients outside the high-probability
 770 region—would contradict the optimality of the minimizer. Combining the two steps yields the stated
 771 upper bound on the inlier misclassification rate ERR_{in} .



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Empirical evidence that InD sample gradients are coordinate-wise sub-Gaussian. (Left) The histogram of gradient values (CIFAR-100 InD data) concentrates around the mean with light tails, as expected under sub-Gaussianity. (Right) The Q-Q plot of empirical InD gradient quantiles against theoretical Gaussian quantiles aligns closely with the reference line, confirming sub-Gaussian behavior. Together these panels justify the sub-Gaussian assumption underlying Theorems 4.1 and 4.2.

Step 1: Concentration of InD Gradients. Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ) of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

787 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

788 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

789 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
790 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

791 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

792 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

793 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

794 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

795 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
796 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
797 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
798 called *good inliers*.

799 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
800 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

801 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
802 thereby contradicting the optimality of $\mathcal{S}^*$.

803 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
804 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

805 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

806 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

807 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

808 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
809 available for swapping in.

810 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
811 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

812 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

813 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

814 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

815 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

816 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
817 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

818 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

819 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

820 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

821 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

822 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

823 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
824 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

825 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
826 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
827 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

828 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

829 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
830 increase it, so the median distance from μ_j does not increase and possibly decreases.

831 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

832 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

833 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
834 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

835 □

836 C.2 Bound on the Outlier Retention Rate

837 We now establish a non-asymptotic upper bound on the fraction of OOD points retained in the subset
838 returned by the MEDIX filter. The result parallels the inlier misclassification bound (Appendix C.1)
839 but, rather than relying on inlier concentration, exploits a separation assumption between the OOD
840 and InD gradient means.

841 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
 842 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

843 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

844 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

845 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

846 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

847 **Proof. Step 1: Concentration of OOD Gradients.**

848 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
 849 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

850 has probability at most $p_{\text{out}}(\varepsilon)$.

851 **Counting the ‘‘ambiguous’’ OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

852 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

853 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

854 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

855 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

856 which holds with probability at least $1 - \delta$.

857 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

858 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

879 **Differences with the original bound.** The original, sub-Gaussian bound carries a decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

880 that vanishes exponentially in the dimension d , rendering the bound extremely tight for large d . The
881 present bound, by contrast, invokes only finite fourth moments and thus exhibits the milder decay
882 term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2},$$

883 which decays as $O(1/d)$ and is correspondingly looser at large d . The trade-off is therefore explicit:
884 the original bound requires sub-Gaussian tails, whereas the new one tolerates heavier tails under a
885 mere finite-fourth-moment condition. Notably, both bounds retain the identical finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

886 which shrinks as m_{in} grows regardless of the tail assumption—so the robustness guarantee persists
887 across both regimes.

888 *Proof.* The proof mirrors the structure of the original Theorem 4.1 and again comprises two steps:
889 (1) bounding the fraction of “bad” InD samples (those far from ∇_{in}) via concentration, now without
890 invoking sub-Gaussianity, and (2) bounding the inlier misclassification rate through a swapping
891 argument. The sole substantive change is that the sub-Gaussian tail bound is replaced by a moment-
892 based bound derived from the finite-fourth-moment assumption.

893 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

894 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
895 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

896 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
897 S_{wild} .

898 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
899 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

900 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

901 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

902 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

903 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

904 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

905 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

906 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

907 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
908 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

909 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
910 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

911 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

912 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

913 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

914 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 915 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

916 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 917 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

918 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 919 thereby contradicting the optimality of $\mathcal{S}^*$.

920 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 921 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

922 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

923 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

924 Define the swap size as

$$m'_\epsilon := \min \{m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

925 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 926 available for swapping in.

927 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 928 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

929 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

930 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

931 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

932 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

933 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 934 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

935 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

936 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

937 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

938 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

939 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

940 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 941 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

942 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 943 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 944 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

945 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

946 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 947 increase it, so the median distance from μ_j does not increase and possibly decreases.

948 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

949 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

950 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 951 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

952 □